

ECO387C - Fall 2024

Problem Set 1 Solutions

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Question 1

a The current value Lagrangian is given by:

$$\mathcal{L} = \sum_{t=0}^T \beta^t \left[\ln(c_t^\ell) + \lambda_t^\ell \left(y_t^\ell + b_{t-1}^\ell (1 + r_{t-1}) - c_t^\ell - b_t^\ell \right) \right] + \beta^T \mu_T^\ell b_T^\ell$$

b The first order conditions of this problem are:

$$\frac{\partial \mathcal{L}}{\partial c_t^\ell} = 0 \implies \frac{1}{c_t^\ell} = \lambda_t^\ell \quad \forall t \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial b_t^\ell} = 0 \implies \lambda_t^\ell = \beta \lambda_{t+1}^\ell (1 + r_t) \quad t = 0, \dots, T-1 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial b_T^\ell} = 0 \implies \lambda_T^\ell = \mu_T^\ell \quad (3)$$

Hence, agent ℓ 's consumption-Euler equation is:

$$\frac{1}{c_t^\ell} = \beta(1 + r_t) \frac{1}{c_{t+1}^\ell}$$

c A competitive equilibrium is a sequence of

(a) consumption allocations $\{c_t^A, c_t^B\}_{t=0}^T$,

(b) bonds $\{b_t^A, b_t^B\}_{t=0}^{T-1}$, and

(c) real interest rates $\{r_t\}_{t=0}^{T-1}$

such that taking the real interest rates as given,

(a) Markets clear, and

(b) both agents' optimization problem is solved.

d From the first order conditions above, we know that:

$$\frac{c_{t+1}^A}{c_t^A} = \frac{c_{t+1}^B}{c_t^B} = \beta(1 + r_t) \quad (4)$$

From the market clearing condition, we know that:

$$\begin{aligned} N^A c_t^A + N^B c_t^B &= N^A y_t^A + N^B y_t^B \\ N^A c_{t+1}^A + N^B c_{t+1}^B &= N^A y_{t+1}^A + N^B y_{t+1}^B \end{aligned}$$

Since we know that $N^A = N^B = N$, then imposing the endowment pattern yields:

$$c_t^A + c_t^B = c_{t+1}^A + c_{t+1}^B = 1$$

All together, this means:

$$\frac{c_{t+1}^\ell}{c_t^\ell} = \frac{1 - c_{t+1}^\ell}{1 - c_t^\ell}, \ell = A, B$$

This means that for both agents, their consumption is equivalent across periods in the optimum. Substituting this into the consumption-Euler equation pins down the real interest rate:

$$1 = \beta(1 + r_t)$$

as required.

e The intertemporal budget constraint is given by:

$$\sum_{t=0}^T \frac{c_t^\ell}{\prod_{j=0}^{t-1} (1 + r_j)} = \sum_{t=0}^T \frac{y_t^\ell}{\prod_{j=0}^{t-1} (1 + r_j)}.$$

Since $1 + r_t = \frac{1}{\beta} \forall t$, and we know that consumption is constant across periods, the intertemporal budget constraint simplifies to:

$$c^\ell \sum_{t=0}^T \beta^t = \sum_{t=0}^T \beta^t y_t^\ell$$

f For the sum on the LHS of the intertemporal budget constraint, use the following manipulation:

$$\begin{aligned} S &= 1 + \beta + \beta^2 + \dots + \beta^T \\ S\beta &= \beta + \beta^2 + \dots + \beta^{T+1} \\ S &= \frac{1 - \beta^{T+1}}{1 - \beta} \end{aligned}$$

For the sum on the RHS of the intertemporal budget constraint, note that T is an odd number. This means that for type A agents, the sum written out is:

$$\sum_{t=0}^T \beta^t y_t^A = 1 + \beta^2 + \beta^4 + \dots + \beta^{T-1}$$

Using the same trick as above, this sum simplifies to:

$$\sum_{t=0}^T \beta^t y_t^A = \frac{1 - \beta^{T+1}}{1 - \beta^2}$$

Substituting everything together results in: $c^A = \frac{1}{1+\beta}$. From the market clearing condition that $c^A + c^B = 1$ every period. Therefore, $c_B = \frac{\beta}{1+\beta}$

g Both agents' FOCs are identical to the previous question. However, the goods market clearing condition in *even* periods is now:

$$\begin{aligned} N^A c_t^A + N^B c_t^B &= N^A \\ N^A c_{t+1}^A + N^B c_{t+1}^B &= N^B \end{aligned}$$

From the consumption-Euler equation, we have:

$$c_{t+1}^\ell = \beta c_t^\ell (1 + r_t)$$

Substituting this into the period $t + 1$ market clearing condition:

$$\beta \left[N^A c_t^A + N^B c_t^B \right] (1 + r_t) = N^B$$

Substituting in the period t market clearing condition simplifies the above to

$$\beta(1 + r_t) = \frac{N^A}{N^B}$$

Letting E and O denote the even and odd periods respectively, the equilibrium interest rates are now:

$$1 + r^E = \frac{N^B}{\beta N^A}, 1 + r^O = \frac{N^A}{\beta N^B} \quad (5)$$

This means that for both types of agents, their consumption-Euler equation is now:

$$c^O = \beta(1 + r^E)c^E$$

To obtain the consumption allocations, substitute the above into the agents' intertemporal budget constraint

$$\begin{aligned} \sum_{t=0}^T \frac{c_t^\ell}{\prod_{j=0}^{t-1} (1 + r_j)} &= c^E + \frac{c^O}{1 + r^E} + \frac{c^E}{(1 + r^E)(1 + r^O)} + \dots \\ &= c^E (1 + \beta) \left[1 + \frac{1}{(1 + r^O)(1 + r^E)} + \frac{1}{((1 + r^O)(1 + r^E))^2} \dots \right] \end{aligned} \quad (6)$$

Upon rearranging (5), we can see that:

$$(1 + r^O)(1 + r^E) = \frac{1}{\beta^2}$$

Substituting this into Equation 6 results in:

$$\sum_{t=0}^T \frac{c_t^\ell}{\prod_{j=0}^{t-1} (1 + r_j)} = c^E (1 + \beta) \left[1 + \beta^2 + \beta^4 + \dots + \beta^{T-1} \right]$$

Using the same steps as in part F results in the LHS of the intertemporal budget constraint being:

$$\sum_{t=0}^T \frac{c_t^\ell}{\prod_{j=0}^{t-1} (1 + r_j)} = c^E (1 + \beta) \frac{1 - \beta^{T+1}}{1 - \beta^2}$$

As in part F, the RHS of the intertemporal budget constraint for type A agents reduces to:

$$\sum_{t=0}^T = \frac{1 - \beta^{T+1}}{1 - \beta^2}$$

Equating the LHS and RHS of Agent A's intertemporal budget constraint in even periods yields:

$$c_E^A = \frac{1}{1 + \beta}$$

We can use Agent A's consumption-Euler equation to find their consumption in odd periods:

$$c_O^A = \frac{N^B}{N^A} \frac{1}{1 + \beta}$$

Imposing market clearing, we can find Agent B's consumption in even and odd periods:

$$c_E^B = \frac{N^A}{N^B} \frac{\beta}{1 + \beta}$$

$$c_O^B = \frac{\beta}{1 + \beta}$$

Question 2

See provided MATLAB code.

For $\rho = 0.95$, the mean of our estimated $\hat{\rho}$'s was 0.90699, indicating a slight downward bias.

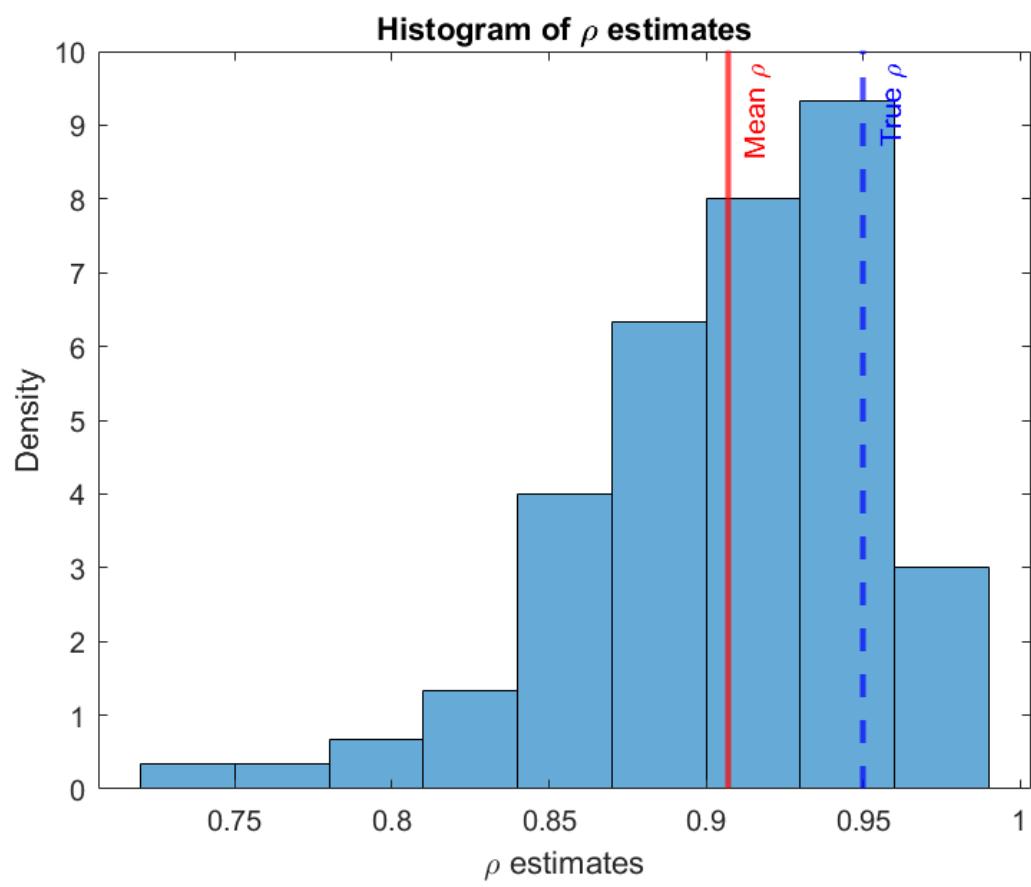


Figure 1: Question 2 Part C.IV